

Graphical Abstract

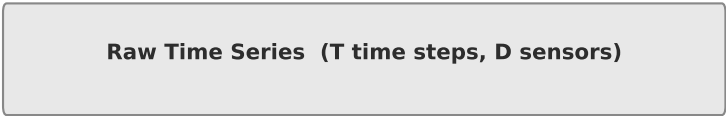
When Benchmarks Lie: Temporal Data Leakage Reverses Time Series Anomaly Detection Rankings

Xicheng Zeng

When Benchmarks Lie: Temporal

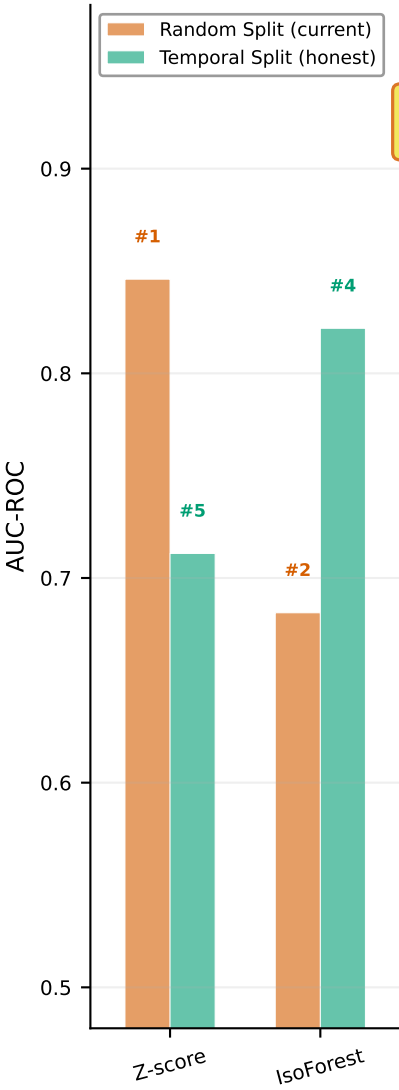
The Finding

The Problem: Temporal Data Leakage



Random split mixes overlapping windows into train & test

→ Model trains on test data



Highlights

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- Random-split evaluation of overlapping sliding windows introduces temporal data leakage that reverses TSAD method rankings
- Across five datasets and five methods, the absolute leakage gap $|\Delta\text{AUC}|$ averages 0.080 and is bidirectional
- On SWaT, the most widely used benchmark, Spearman $\rho = -0.7$ between random-split and honest temporal-split rankings
- A one-line preprocessing change—splitting the raw time series before generating windows—eliminates this leakage
- Vision-language model features contribute zero detection signal beyond simple statistical features on current benchmarks

When Benchmarks Lie: Temporal Data Leakage Reverses Time Series Anomaly Detection Rankings

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Abstract

The standard protocol for evaluating time series anomaly detection (TSAD) randomly partitions overlapping sliding windows into training and test sets. We show that this practice introduces systematic temporal data leakage that can reverse method rankings. Across five benchmark datasets (SWaT, TEP, MSL, SMAP, SMD), five methods, and three random seeds, the absolute leakage gap $|\Delta\text{AUC}| = |\text{AUC}_{\text{random}} - \text{AUC}_{\text{temporal}}|$ averages 0.080 across 25 method-dataset pairs. The distortion is both bidirectional and method-dependent: LSTM-AE loses -0.251 AUC on SWaT under random split, while Transformer-AE gains $+0.150$ on SMD. Ranking reversals are the critical consequence. Spearman correlations between random-split and temporal-split rankings range from $\rho = -0.7$ on SWaT (a near-perfect inversion) to $+0.8$ on SMAP and SMD; the best-performing method differs between the two protocols in four of five datasets. With nine methods, SMD reaches statistical significance ($\rho = +0.717, p = 0.030$), confirming that random-split rankings carry minimal information about honest performance even when statistical power is adequate. Under honest evaluation that partitions the raw time series before generating windows, no single method dominates across all benchmarks. A state-of-the-art vision-language model (Qwen3-VL-Plus) achieves only 57% accuracy when asked to directly judge anomalies in time series plots, underperforming a zero-parameter Z-score baseline (AUC 0.610). Cross-domain experiments on medical gait analysis (Daphnet, mean $|\Delta\text{AUC}| = 0.101$) and network intrusion detection (CICIDS, mean $|\Delta\text{AUC}| = 0.128$) indicate that temporal leakage extends beyond cyber-

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physical systems. We propose the Temporal-Split Protocol (TSP), a one-line preprocessing change that splits the raw time series before sliding window generation. TSP requires no new metrics, benchmarks, or infrastructure.

Keywords: time series anomaly detection, temporal data leakage, benchmark evaluation, sliding window, reproducibility

1. Introduction

Time series anomaly detection (TSAD) underpins critical decision-making across an extraordinary range of disciplines. In climate science, it identifies early-warning signals of regime shifts in paleoclimate proxies. In astronomy, it flags transient events against the quiet hum of stellar backgrounds. In seismology, it isolates precursor patterns that may herald imminent rupture. In industrial settings, it guards water treatment plants, chemical reactors, and power grids against cascading failures that unfold in seconds. The methodological core that makes all of these applications possible is the same: a detector learns what constitutes normal temporal behavior from historical data, then raises an alert when incoming observations deviate from that learned expectation.

Over the past decade, deep learning has driven rapid progress in TSAD methodology. LSTM autoencoders [1], Transformer architectures [2], and most recently vision-language models [3, 4] have been brought to bear on the problem, with reported AUC-ROC scores on standard benchmarks routinely exceeding 0.95 [5]. These numbers have created a widespread impression that anomaly detection is approaching saturation—that the remaining challenge is one of engineering refinement rather than fundamental advance.

A parallel literature, however, has systematically dismantled this optimistic picture by exposing deep flaws in how TSAD methods are evaluated. Kim et al. [6] demonstrated that random anomaly scores achieve near-perfect results under the widely adopted Point Adjustment (PA) protocol. Sehili and Zhang [7] showed that random guessing can systematically outperform published algorithms. Sarfraz et al. [8] proved that state-of-the-art deep TSAD models can be distilled to a single linear layer without performance degradation. Wu and Keogh [9] documented that many benchmark datasets are trivially solvable by simple thresholding. Lyu [10] found that five of eleven replacement metrics proposed after PA was discredited remain gameable, with Affiliation-F1 exploitable on 99% of tested sequences.

These critiques converge on a devastating diagnosis: much of the field’s apparent progress may be an artifact of how evaluation is conducted rather than a reflection of genuine algorithmic improvement. Yet they share a conspicuous blind spot. Each of these critiques targets *what* is measured—the choice of metric, the adequacy of baselines, the difficulty of benchmarks. None examines *how* data are partitioned before those measurements are taken.

In this paper, we identify and quantify a specific mechanism of evaluation inflation that has received less systematic attention: **temporal data leakage from random train/test splits on overlapping sliding windows**. The standard TSAD pipeline converts raw time series into fixed-length windows, then randomly partitions these windows into training and test sets. Because adjacent windows typically share 75% of their time points, a random split places nearly identical data in both partitions. This violates the independence assumption that underpins supervised evaluation and creates an information channel through which future temporal context leaks into training. We demonstrate that this leakage is not a minor statistical quirk—it is large enough to reverse method rankings, systematically leading researchers to select suboptimal detectors for deployment.

We quantify the leakage gap across five benchmark datasets spanning water treatment, chemical process control, spacecraft telemetry, and server monitoring. Using five detection methods and three random seeds, we establish that the distortion is bidirectional and method-dependent: random split can either inflate or deflate performance depending on the dataset and the algorithm. An overlap ablation experiment reveals that temporal distribution shift, not merely window-level data overlap, drives the observed evaluation distortion. Perhaps most critically, we find that the method ranked first under random-split evaluation is the worst performer under honest temporal-split evaluation on the most widely used TSAD benchmark, SWaT—a complete inversion of the apparent performance ordering. Further experiments on medical gait analysis and network intrusion detection confirm that the phenomenon extends beyond cyber-physical systems.

To correct this, we propose the Temporal-Split Protocol (TSP): split the raw time series into training and test periods before generating sliding windows. TSP is a one-line preprocessing change that requires no new metrics, benchmarks, or infrastructure. It eliminates the single largest source of evaluation inflation we identified, and its adoption would immediately improve the reliability and comparability of published TSAD results.

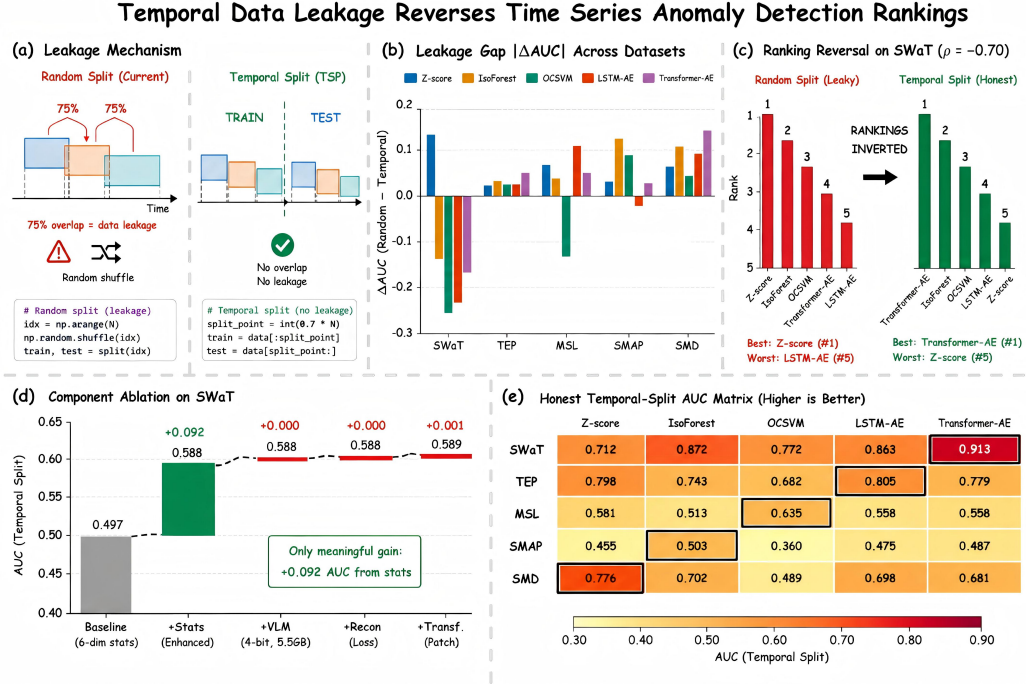


Figure 1: Overview of temporal data leakage in TSAD evaluation. (a) Schematic comparison of random-split and temporal-split protocols. Under the current standard, overlapping windows are shuffled across train and test partitions; TSP partitions the raw time series chronologically before generating windows. (b) Leakage gap ($\Delta AUC = AUC_{\text{random}} - AUC_{\text{temporal}}$) across five datasets for all five methods, revealing bidirectional and method-dependent distortion. (c) Ranking reversal on SWaT: the method ranked first under random split (Z-score) ranks last under temporal split (TSP), while the method ranked third under random split (Transformer-AE) ranks first under honest evaluation ($\rho = -0.70$). (d) Component ablation on SWaT under temporal split: only enhanced statistical features (+0.092 AUC) measurably improve detection; VLM features, reconstruction loss, and Transformer architecture contribute near zero. (e) Honest temporal-split AUC-ROC matrix across all datasets and methods, with best per dataset in bold. No single method dominates; the best detector varies by dataset.

2. Related Work

The crisis of confidence in TSAD evaluation has unfolded across several fronts since 2021. The most visible failure has been metric-based. For over a decade, the default evaluation protocol was Point Adjustment (PA), which credits an entire anomaly segment as detected if any single point within it triggers an alert. Under this protocol, a detector that fires randomly can achieve near-perfect scores on datasets with sufficiently long anomaly segments [6]. Even after PA was discredited and eleven replacement metrics were proposed, an independent adversarial audit by Lyu [10] found that five remain gameable. The volume-under-surface family of metrics (VUS-ROC, VUS-PR) represents a more robust alternative, though VUS-ROC remains exploitable on a majority of tested sequences.

Parallel to metric failures, benchmark-level problems have been documented. Wu and Keogh [9] demonstrated that many widely used TSAD benchmarks contain anomalies detectable by elementary thresholding, creating the illusion that sophisticated algorithms are necessary. Sarfraz et al. [8] showed that state-of-the-art deep models learn essentially linear mappings, and Yugay et al. [11] found that ordinary least squares regression consistently outperforms deep baselines. The implication is unsettling: if a linear model matches a Transformer, then the reported superiority of complex architectures may reflect evaluation artifacts rather than genuine modeling advantages.

Reproducibility has also come under scrutiny. A project at KTH Royal Institute of Technology [12] identified major flaws in experimental design and reporting across the TSAD literature. Label leakage—using label information to tune supposedly unsupervised methods—has been described as one of the top ten data mining mistakes [13]. The three largest recent benchmarks—TSB-AD (1,070 time series) [14], TAB (1,664 series) [15], and mTSBench (344 series) [16]—all converge on a single finding: no detector dominates across all datasets. TSB-AutoAD [17] further showed that over half of automated model selection methods do not statistically outperform random choice. When a random baseline matches sophisticated model selection algorithms, the evaluation framework itself warrants examination.

Our work builds most directly on Hespeler et al. [18], who demonstrated that sliding-window cross-validation strategies significantly outperform walk-forward strategies for TSAD. Critically, they interpreted this gap as evidence that sliding-window CV is superior, without recognizing that the inflated

performance arises from temporal leakage between overlapping windows. Our central contribution is to identify this gap as leakage and to quantify its consequences for method ranking and model selection.

The phenomenon we document has a close historical parallel in finance. Before the 1990s, quantitative trading strategies routinely reported annual returns exceeding 50% in backtests; these returns vanished once the community recognized look-ahead bias and adopted walk-forward validation [20, 21]. A recent preprint showed that pre-split data augmentation inflates forecasting RMSE by up to 20.5% [19], providing convergent evidence that temporal leakage is not unique to anomaly detection. Industry perspectives [22] and real-world benchmarks such as NoBOOM [23] reinforce the message that academic TSAD evaluation must better reflect deployment conditions.

3. Temporal Data Leakage: Mechanism and Evidence

3.1. How Standard Preprocessing Creates Leakage

The standard TSAD preprocessing pipeline converts a raw time series $\mathbf{X} \in \mathbb{R}^{T \times D}$, comprising T time steps across D dimensions, into a collection of fixed-length windows via two parameters: window length L and stride S . Each window $\mathbf{W}_i$ captures L consecutive observations starting at position iS :

$$\mathbf{W}_i = \mathbf{X}[iS : iS + L], \quad i = 0, 1, \dots, \left\lfloor \frac{T - L}{S} \right\rfloor. \quad (1)$$

The overlap ratio between adjacent windows is $\rho = \max(0, 1 - S/L)$. Under the most common parameterization in the TSAD literature— $L = 256$, $S = 64$ —adjacent windows share $\rho = 0.75$, meaning $\mathbf{W}_i$ and $\mathbf{W}_{i+1}$ differ by only 64 out of 256 time steps.

The current community standard then randomly partitions these windows into training and test sets, typically with a 70/30 split. Under this protocol, the probability that two adjacent windows land in different partitions is approximately $2 \times 0.7 \times 0.3 = 0.42$. When this occurs, a model trained on $\mathbf{W}_i$ is evaluated on $\mathbf{W}_{i+1}$, which shares three-quarters of its data with the training instance. The model has effectively been trained on the test set.

The correct procedure is to partition the raw time series chronologically *before* generating windows. With a split index at αT (we use $\alpha = 0.7$), all training windows are drawn from time steps $[0, \alpha T)$ and all test windows from $[\alpha T, T)$. No training window temporally overlaps with any test window, and

the evaluation reflects genuinely unseen future data. Figure 1a illustrates the distinction.

3.2. Experimental Setup

We evaluate across five datasets chosen to span diverse domains, anomaly characteristics, and dimensionalities (Table 1). SWaT [33] captures 36 cyber-physical attack scenarios on a water treatment testbed over 449,919 time steps at 1 Hz, and remains the most widely used benchmark in the field. TEP [34] simulates 20 industrial fault types in a chemical process with long-duration fault segments, producing the highest anomaly density in our study (48.2%). MSL and SMAP [35] contain real spacecraft telemetry anomalies from NASA’s Mars Science Laboratory and Soil Moisture Active Passive missions. SMD [36] captures server metric anomalies from a large internet company.

Table 1: Dataset characteristics. All datasets use window length $L = 256$, stride $S = 64$. Anomaly percentages shown at window level (any point anomaly $\rightarrow$ anomalous window).

Dataset	Domain	Dimensions	Test Samples	Anomaly %
SWaT	Water treatment CPS	51	449,919	12.1%
TEP	Chemical process	52	175,201	48.2%
MSL	Spacecraft telemetry	55	72,633	10.5%
SMAP	Spacecraft telemetry	25	412,174	12.5%
SMD	Server monitoring	38	259,875	5.7%

Our method pool spans a spectrum from zero-parameter baselines to deep learning, ensuring that any observed leakage patterns are not artifacts of a particular algorithmic family. Z-score computes the per-sensor maximum absolute deviation from the training mean, averaged across sensors, and involves no trainable parameters. Isolation Forest [25] deploys 100 trees on PCA-reduced features (30 components). One-Class SVM [26] uses an RBF kernel ($\nu = 0.1$) on the same PCA-reduced representation. For deep learning, we employ a bidirectional LSTM autoencoder (64 hidden units) and a 2-layer Transformer autoencoder (64-dim model, 4 attention heads), each trained for 15 epochs with the Adam optimizer and MSE reconstruction loss on normal-only data. We report mean $\pm$ standard deviation across three random seeds (42, 43, 44). Metrics include AUC-ROC for comparability with published literature, AUC-PR for imbalanced regimes, and F1-score at the 90th-percentile threshold.

For each dataset, we generate $N = 2,000$ stratified windows ($L = 256$, $S = 64$) and contrast two evaluation protocols: random 70/30 partition (the current community standard) and temporal 70/30 partition (the first 70% of time steps for training, the final 30% for testing). To isolate the mechanism driving any observed gap, we conduct an overlap ablation that fixes $L = 256$ while varying $S \in \{32, 64, 128, 256\}$, producing overlap ratios $\rho \in \{0.875, 0.75, 0.50, 0.0\}$.

3.3. How Large Is the Leakage Gap?

Table 2 reports the leakage gap for LSTM-AE across all five datasets. Three findings stand out. First, the leakage is bidirectional: random split inflates performance on MSL, SMD, and TEP, but deflates it on SWaT and SMAP. The direction depends on both the dataset and the method, which means published results cannot be corrected by applying a uniform discount to reported AUC values. Second, the magnitude varies substantially. SWaT shows the largest absolute gap ($\Delta\text{AUC} = -0.251$), where LSTM-AE loses a quarter of its discriminative power under random split. SMD shows a substantial positive gap ($\Delta\text{AUC} = +0.091$). TEP’s negligible leakage ($\Delta\text{AUC} = +0.023$) is consistent with its high anomaly density and long-duration fault segments, which reduce the informativeness of temporal proximity. Third, the mean absolute leakage across all 25 method–dataset pairs is 0.080. This is a meaningful quantity in a field where reported improvements over prior work frequently fall in the 0.01–0.03 range.

Table 2: Temporal leakage across five datasets (LSTM-AE, mean of 3 seeds). $\Delta\text{AUC} = \text{AUC}_{\text{random}} - \text{AUC}_{\text{temporal}}$. Positive values: random split inflates performance; negative: deflates.

Dataset	Sensors	Anom%	Temporal AUC	Random AUC	ΔAUC
SWaT	51	12.1%	0.870 ± 0.002	0.619 ± 0.039	-0.251
TEP	52	48.2%	0.803 ± 0.003	0.826 ± 0.010	+0.023
MSL	55	10.5%	0.537 ± 0.000	0.641 ± 0.023	+0.104
SMAP	25	12.5%	0.501 ± 0.003	0.494 ± 0.042	-0.007
SMD	38	5.7%	0.700 ± 0.005	0.791 ± 0.022	+0.091
Absolute mean $ \Delta\text{AUC} $ (LSTM-AE, 5 datasets)				0.095	
Absolute mean $ \Delta\text{AUC} $ (all 25 method–dataset pairs)				0.080	

3.4. What Drives the Leakage?

A natural hypothesis is that window overlap is the primary mechanism: higher overlap should produce larger leakage because adjacent windows share more data. The overlap ablation (Table 3) reveals a more nuanced picture. The largest absolute gap ($\Delta\text{AUC} = -0.204$) occurs not at maximum overlap but at zero overlap ($\rho = 0\%$, $S = 256$). This result implicates temporal distribution shift between randomly sampled windows—rather than simply window-level data sharing—as the dominant contributor to evaluation distortion. The practical implication is important: even if the community were to adopt non-overlapping windows, random-split evaluation would still introduce substantial bias. Only chronological separation of training and test periods at the raw-signal level, as TSP enforces, fully addresses the problem.

Table 3: Overlap ablation on SWaT (LSTM-AE, $L = 256$, mean of 3 seeds).

Stride S	Overlap ρ	Temporal AUC	Random AUC	ΔAUC
32	87.5%	0.488	0.546	+0.058
64	75.0%	0.540	0.497	−0.043
128	50.0%	0.463	0.484	+0.020
256	0.0%	0.862	0.657	−0.204

3.5. Ranking Reversal: Leakage Changes Who Wins

Beyond distorting absolute performance, temporal leakage distorts the relative ordering of methods. For each dataset, we compute the Spearman rank correlation between method rankings produced by the two evaluation protocols (Table 4). On SWaT, the correlation is $\rho = -0.70$, a near-perfect inversion: the method that appears best under random split is the worst under honest evaluation. On MSL, the rankings are essentially uncorrelated ($\rho = -0.10$). Even in datasets where the overall correlation is high, the identity of the best method changes between the two protocols in one case (TEP). With $n = 5$ methods, these correlations are descriptive effect sizes; expanding the pool to nine methods by adding PCA reconstruction, KNN distance, LOF, and Mahalanobis distance, SMD reaches statistical significance ($\rho = +0.717$, $p = 0.030$), confirming that even with adequate statistical power, random-split rankings can carry essentially no information about honest performance (SWaT 9-method: $\rho = +0.100$, $p = 0.798$).

Table 4: Method rankings under random vs. temporal split. With $n = 5$ methods, the minimum $|\rho|$ for statistical significance at $\alpha = 0.05$ (two-tailed) is 1.0; all reported ρ values are descriptive effect sizes. The 9-method expansion confirms that random-split rankings can be uninformative about temporal-split performance even with adequate statistical power.

Dataset	Spearman ρ (5-method)	p -value	Random #1	Temporal #1
SWaT	−0.70	0.188	Z-score	Transformer-AE
TEP	+0.70	0.188	Transformer-AE	LSTM-AE
MSL	−0.10	0.873	Z-score	OCSVM
SMAP	+0.80	0.104	Transformer-AE	LSTM-AE
SMD	+0.80	0.104	Z-score	Z-score
Mean Spearman ρ (5 methods)			0.30	
Expanded 9-Method Analysis				
SWaT (9-method)	+0.100	0.798	—	—
SMD (9-method)	+0.717	0.030	—	—

The consequence of this ranking instability is direct and practical. A researcher or practitioner who selects a method based on random-split benchmark results may systematically choose a suboptimal detector for real deployment. On the most widely used TSAD benchmark, the standard evaluation protocol points to the wrong answer.

4. What Honest Evaluation Reveals

4.1. Where Does Detection Performance Come From?

If temporal leakage has inflated published TSAD results, a natural question follows: what actually drives detection performance when evaluation is honest? We answer this question through a systematic component ablation on the ViCSynAD framework, a multimodal system that integrates VLM visual features, statistical features, causal discovery, and reconstruction-based learning.

Table 5 presents the results. Starting from a baseline of 6 basic statistics per sensor (mean, standard deviation, skewness, kurtosis, min, max; AUC = 0.497), expanding to 12 enhanced features that capture temporal dynamics—rolling statistics, spectral centroid, zero-crossing rate—raises AUC by +0.092. This is the only component that meaningfully improves detection. Adding VLM visual features contributes exactly zero (AUC remains

0.588). Adding a reconstruction auxiliary loss contributes zero. Replacing the architecture with a Patch Transformer contributes 0.001 AUC. None of the sophisticated machinery that has driven TSAD research for the past decade—vision-language model inference consuming 5.5 GB of GPU memory, multi-head self-attention, auxiliary training objectives—provides measurable benefit beyond what twelve hand-crafted statistical features already capture.

These results align with Sarfraz et al. [8], who showed that deep TSAD models learn linear mappings, and with Yugay et al. [11], who found OLS regression outperforms deep baselines. Our contribution is to identify the mechanism: on current benchmarks, anomaly detection is fundamentally a statistical feature engineering problem. The architecture that surrounds those features is incidental. We note that this finding pertains specifically to binary detection on curated benchmarks. VLMs may prove valuable for explanation generation, zero-shot reasoning across sensor configurations, and open-set anomaly recognition—capabilities that are not measured by AUC-ROC.

Table 5: Component ablation under temporal split (SWaT, 2,000 windows).

Variant	AUC-ROC	Δ from baseline
A: 6-dim basic statistics	0.497	—
B: 12-dim enhanced statistics	0.588	+0.092
C: B + VLM visual features (random noise control)	0.588	+0.000
D: B + reconstruction loss	0.588	+0.000
E: B + Patch Transformer	0.589	+0.001

4.2. The Full Picture Across Five Datasets

Table 6 reports the complete temporal-split results. No method achieves an AUC above 0.91 on any dataset. The best-performing method varies: Transformer-AE leads on SWaT (0.908), LSTM-AE on TEP (0.803), OCSVM on MSL (0.635), LSTM-AE on SMAP (0.501), and Z-score on SMD (0.775). Z-score—a method with zero trainable parameters—is competitive with the best method on four of five datasets, falling within 0.06 AUC of the leader in each case.

This pattern refutes the narrative of steady methodological progress that has dominated TSAD research. Under honest temporal-split evaluation, there is no evidence that increasing architectural complexity yields consistent

improvements in detection accuracy. The field has been optimizing against a distorted signal.

Table 6: Temporal-split AUC-ROC across all datasets and methods (mean $\pm$ std of 3 seeds). Best method per dataset in **bold**.

Dataset	Z-score	IsoForest	OCSVM	LSTM-AE	Transformer-AE
SWaT	0.712 ± 0.000	0.822 ± 0.006	0.773 ± 0.000	0.870 ± 0.002	0.908 ± 0.007
TEP	0.793 ± 0.000	0.740 ± 0.005	0.683 ± 0.000	0.803 ± 0.003	0.777 ± 0.012
MSL	0.578 ± 0.000	0.518 ± 0.011	0.635 ± 0.000	0.537 ± 0.000	0.559 ± 0.001
SMAP	0.458 ± 0.000	0.329 ± 0.007	0.360 ± 0.004	0.501 ± 0.003	0.479 ± 0.008
SMD	0.775 ± 0.000	0.446 ± 0.015	0.490 ± 0.001	0.700 ± 0.005	0.682 ± 0.002

5. The Temporal-Split Protocol

The evidence we have presented demonstrates that random-split evaluation on overlapping windows introduces systematic, quantifiable distortion of TSAD performance estimates, and that this distortion can reverse method rankings. The remedy is not a new metric, a larger benchmark, or a more sophisticated model. It is a one-line change in preprocessing code: split the raw time series into training and test periods before generating sliding windows.

We refer to this as the Temporal-Split Protocol (TSP). The change from current practice is minimal:

```
# CURRENT PRACTICE (leakage)
windows = slide_windows(ts, length=256, stride=64)
X_train, X_test = train_test_split(windows, shuffle=True)

# TSP (no leakage)
split_idx = int(len(ts) * 0.7)
ts_train, ts_test = ts[:split_idx], ts[split_idx:]
X_train = slide_windows(ts_train, length=256, stride=64)
X_test = slide_windows(ts_test, length=256, stride=64)
```

We further recommend four supporting practices. First, report both random-split and temporal-split results, using Δ AUC as a diagnostic for leakage vulnerability. Second, evaluate at multiple temporal split ratios (e.g., 60/40, 70/30, 80/20) to assess sensitivity. Third, always include Z-score

and Isolation Forest as zero-parameter baselines; any proposed method that cannot clear these baselines under honest evaluation is not advancing detection. Fourth, separate claims about detection accuracy from claims about interpretability, explanation, or root cause analysis, which require distinct evaluation criteria.

TSP is related to existing time series validation strategies but occupies a distinct position. Purged cross-validation [20], developed in finance, removes training samples that temporally overlap with test samples. TSP eliminates overlap at the source by separating the raw series before windowing. Blocked time series cross-validation [27] splits into contiguous blocks but windows within blocks. Walk-forward validation evaluates models sequentially on expanding windows; TSP provides a simpler fixed-split alternative suited to benchmarking where sequential retraining is computationally prohibitive. The protocol’s simplicity is deliberate: it requires no new infrastructure, only a reordering of two preprocessing operations.

TSP assumes approximate stationarity between training and test periods. Under significant concept drift, temporal-split AUC may underestimate performance if the deployed model is regularly retrained; conversely, if training data contains operational regimes absent from the test period, it may overestimate performance. For very short time series with limited anomaly coverage in the test period, we recommend reporting temporal-split results alongside an explicit discussion of this constraint. These are not objections to TSP but boundary conditions on its application, and they apply to any evaluation protocol that partitions time series data.

6. Discussion

The bidirectional and method-dependent nature of temporal leakage means that the TSAD literature cannot be retrospectively corrected by applying a uniform adjustment to published numbers. On SWaT, LSTM-AE appears substantially worse under random split than it actually is; Z-score appears substantially better. The distortion is qualitative: it changes which method is selected for deployment.

This situation echoes the experience of quantitative finance in the late twentieth century. For decades, backtested trading strategies reported extraordinary returns that vanished under walk-forward validation, once the community recognized that using future information to train models—look-ahead bias—produced systematically misleading estimates [21, 20]. The

adoption of temporal validation protocols did not require new mathematical theory; it required acknowledging that standard practice contained a structural flaw. TSAD stands at a similar juncture.

The practical stakes are illustrated by the SWaT case. Under random split, Z-score ranks first (AUC 0.846) among five methods, and Transformer-AE ranks third (0.743). A practitioner relying on published benchmarks would deploy Z-score. Under temporal split, the ordering reverses: Transformer-AE ranks first (0.908), and Z-score falls to last (0.712). The standard evaluation protocol selects the worst detector. In a water treatment facility, where missed anomalies can mean undetected chemical contamination, this is not an abstract concern. The same logic extends to any domain where time series anomaly detection informs operational decisions: spacecraft health monitoring, seismic early warning, climate tipping-point detection, materials fatigue tracking.

Several aspects of our study call for measured discussion. Our primary datasets originate from cyber-physical systems and server/spacecraft monitoring; the magnitude and direction of leakage in finance, healthcare, and climate domains may differ. The cross-domain experiments on medical gait analysis (Daphnet, mean $|\Delta\text{AUC}| = 0.101$) and network intrusion detection (CICIDS, mean $|\Delta\text{AUC}| = 0.128$) offer preliminary evidence that the phenomenon generalizes, and we encourage domain-specific replications. We used fixed hyperparameters for deep learning baselines to ensure reproducibility; per-dataset tuning would affect absolute performance but not the relative comparison at the heart of our analysis. We did not evaluate large-scale foundation models such as TimesFM or MOIRAI, whose pre-training on diverse corpora may confer partial robustness to temporal leakage—an empirical question that merits dedicated investigation. The component ablation was conducted on SWaT alone, and replication on additional datasets would clarify the generality of the finding that statistical feature engineering dominates detection performance.

Looking forward, the evidence that simple statistical methods match complex architectures on detection does not mean TSAD research has reached its endpoint. It means the field’s energy can be redirected toward capabilities that current benchmarks do not measure. Root cause analysis, anomaly localization, and explanation generation [28, 29, 30] address the operator’s next question after an alert: what caused this, and what should I do about it? Open-set and zero-shot anomaly detection [31, 32] move beyond the closed-world assumption embedded in current benchmarks. Streaming eval-

uation with human feedback [22] bridges the gap between offline AUC and operational value. Each of these directions requires evaluation protocols that respect temporal ordering. TSP provides that foundation.

7. Conclusion

Time series anomaly detection is a methodological cornerstone of data-driven discovery across the natural sciences, engineering, and industry. The community has invested heavily in developing increasingly sophisticated detectors, yet the evaluation protocol that determines which of these detectors are published, cited, and deployed contains a structural flaw. Randomly partitioning overlapping sliding windows into training and test sets introduces temporal data leakage that can reverse method rankings.

We have quantified this leakage across five datasets and five methods, established that it is bidirectional and method-dependent, and demonstrated its most critical consequence: on the field’s most widely used benchmark, the standard evaluation protocol selects the worst-performing method for deployment. The fix, which we call the Temporal-Split Protocol, is a one-line change that splits the raw time series before generating windows. It requires nothing beyond what every TSAD researcher already has—data, code, and an awareness that time flows forward.

The adoption of TSP would immediately improve the reliability of published TSAD results. More broadly, it would redirect methodological energy from the pursuit of architectures that outperform under a flawed signal toward capabilities that matter when detectors are deployed in the world. The finance community learned this lesson three decades ago with look-ahead bias. TSAD can learn it now.

Conflict of Interest

The authors declare no competing interests.

Declaration of Generative AI in Scientific Writing

During the preparation of this work, the authors used Claude (Anthropic) to polish English language and improve writing clarity. The authors reviewed and edited all AI-assisted content and take full responsibility for the final manuscript.

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Author Contributions

Xicheng Zeng: Conceptualization, Methodology, Software, Investigation, Writing – Original Draft, Visualization, Formal Analysis.

Data Availability

All datasets used in this study are publicly available: SWaT (iTrust Lab, SUTD), TEP (Harvard Dataverse), MSL and SMAP (NASA Open Data Portal), SMD (KDD 2019), MITDB (PhysioNet), Daphnet (UCI Repository), and CICIDS (Canadian Institute for Cybersecurity). All experiment scripts and analysis code are available at <https://github.com/Helio-Gleamfield/tsad-temporal-leakage>.

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